

ALLOSTERIC COUPLING BETWEEN A LIPID BILAYER AND A MEMBRANE PROTEIN

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ABSTRACT - Biological membranes are highly complex and crowded environments that are essential for cellular function, as they mediate molecular transport and regulate key processes such as signaling. They consist primarily of a lipid bilayer associated with membrane proteins, forming a dynamic structure that ensures proper regulation of membrane activities. Yet, to what extent do lipid dynamics regulate membrane protein conformational dynamics and how do membrane proteins locally modulate lipid dynamics is largely unknown. In this work, we use liquid-state ¹H and ¹³C NMR under variable temperature and hydrostatic pressure, combined with molecular dynamics simulations, to elucidate at the atomic scale the allosteric coupling between the DMPC lipid bilayer and the model membrane protein OmpX in nanodiscs [1,2]. We discovered that modifying the mechanical properties of the lipid bilayer directly influences the energy landscape of amino acid side chains at the lipid/protein interface. The dynamics and chemical environment of these residues are directly coupled to the DMPC temperature or pressure-induced phase transition. Unexpectedly, side chains located in the protein core, away from the lipids, are also affected by the change in lipid dynamics during the fluid/rigid phase transition. These findings suggest the existence of a thermodynamically coupled, yet kinetically decoupled, allosteric pathway between lipid dynamics and the internal conformation of membrane proteins, directly impacting our understanding of membrane function. We also found that OmpX delays the lipid stiffening

process by liquefying the annular shell of lipids at the protein-lipid interface. This perturbation is linked to the roughness of the transmembrane surface of OmpX and to a mismatch between the height of its hydrophobic region and the thickness of the rigid phase membrane, creating a hydrophobic mismatch that locally promotes lipid fluidity. This approach helps to better understand how cells may regulate membrane protein function through local modulation of lipid phase behavior.

REFERENCES

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